Python Coding Interview Notes

1. Swap Two Numbers

```
a=10
b=20
a,b=b,a
print(a,b)
```

Output:

```
20 10
```

2. Find Data Type of a Variable

```
a=10
print(type(a))
```

Output:

```
<class 'int'>
```

3. Take User Input

```
name=input("Enter your name: ")
print("Hello",name)
```

Output:

```
Enter your name: Keerti
Hello Keerti
```

4. Calculate Area of a Circle

```
r=float(input())
area=3.14rr
print(area)
```

Input:

```
5
```

Output:

```
78.5
```

5. Find Maximum of Three Numbers

```
a=int(input())
b=int(input())
c=int(input())
if a>b and a>c:
    print(a)
elif b>c:
    print(b)
else:
    print(c)
```

Input:

```
10
25
15
```

Output:

```
25
```

6. Reverse a String (Using Slicing)

```
s=input()
print(s[::-1])
```

Input:

```
Python
```

Output:

```
nohtyP
```

7. Reverse a String Without Using Slicing

```
s=input()
rev=""
for ch in s:
    rev=ch+rev
print(rev)
```

Input:

```
Python
```

Output:

```
nohtyP
```

8. Check Palindrome String (Using Slicing)

```
s=input()
if s==s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```
Input:
madam
Output:
Palindrome

9. Check Palindrome Without Using Slicing

```
s=input()
rev=""
for ch in s:
    rev=ch+rev
if s==rev:
    print("Palindrome")
else:
    print("Not Palindrome")
```
Input:
level
Output:
Palindrome

10. Count Vowels

```
s=input().lower()
count=0
for ch in s:
    if ch in "aeiou":
        count+=1
print(count)
```
Input:
Python Programming
Output:
4

11. Count Uppercase and Lowercase Letters

```
s=input()
upper=0
lower=0
for ch in s:
      if ch.isupper():
          upper+=1
     elif ch.islower():
         lower+=1
print("Uppercase:",upper)
print("Lowercase:",lower)
```
Input:
PyThOn
Output:
Uppercase: 3
Lowercase: 3

12. Remove Spaces

```
s=input()
print(s.replace(" ",""))
```
Input:
Python Programming
Output:
PythonProgramming

13. Count Words in a String

```
s=input()
words=s.split()
print(len(words))
```
Input:
Python is easy to learn

Output:
5

14. Find Duplicate Characters

```
s=input()
seen=""
for ch in s:
   if s.count(ch)>1 and ch not in seen:
        print(ch,end=" ")
        seen+=ch
```

Input:
programming
Output:
r g m

15. Character Frequency

Method 1:

```
s=input()
ch=input()
count=0
for i in s:
    If i==ch:
        count+=1
print(count)
```

Input:
programming
g
Output:
2

Method 2:

```
s=input()

ch=input()

print(s.count(ch))
```

16. Check Anagram

```
s1=input()
s2=input()
if sorted(s1)==sorted(s2):
     print("Anagram")
else:
     print("Not Anagram")
```

Input:
listen
silent
Output:
Anagram

17. Find Largest Element

Method 1:

```
arr=[10,20,5,40,15]
largest=arr[0]
for num in arr:
    if num>largest:
        largest=num
print(largest)
```

Output:
40

Method 2:

```
arr=[10,20,5,40,15]

print(max(arr))
```

18. Find Smallest Element
Method 1:

```
arr=[10,20,5,40,15]
smallest=arr[0]
for num in arr:
```

```
    if num<smallest:
         smallest=num
print(smallest)
```

Output:

5

Method 2:

```
arr=[10,20,5,40,15]

print(min(arr))
```

19. Find Second Largest Element

Method 1:

```
arr=[10,20,5,40,15]
largest=max(arr)
second=arr[0]
for num in arr:
     if num>second and num!=largest:
           second=num
 print(second)
```

Output:

20

Method 2:

```
arr=[10,20,5,40,15]

arr.sort()

print(arr[-2])
```

20. Find Largest and Second Largest Without Using Built-in Functions

Method 1:

```
arr=[10,20,5,40,15]
largest=arr[0]
second=arr[0]
for num in arr:
     if num>largest:
          second=largest
          largest=num
    elif num>second and
num!=largest:
         second=num
print("Largest:",largest)
print("Second Largest:",second)
```

Output:

Largest: 40

Second Largest: 20

Method 2:

```
arr=[10,20,5,40,15]
arr.sort()
print("Largest:",arr[-1])
print("Second Largest:",arr[-2]
```

21. Find Sum of Array Elements

Method 1:

```
arr=[1,2,3,4,5]
print(sum(arr))
```

Output:

15

Method 2:

```
arr=[1,2,3,4,5]
total=0
for num in arr:
      total+=num
print(total)
```

22. Remove Duplicates from a List

Method 1:

```
arr=[10,20,10,30,20,40,50,40]
print(list(set(arr)))
```

Output:

[10, 20, 30, 40, 50]

Method 2:

```
arr=[10,20,10,30,20,40,50,40]
new=[]
for num in arr:
    if num not in new:
        new.append(num)
print(new)
```

23. Reverse a List

Method 1:

```
arr=[1,2,3,4,5]
print(arr[::-1])
```

Output:

[5, 4, 3, 2, 1]

Method 2:

```
arr=[1,2,3,4,5]
arr.reverse()
print(arr)
```

24. Sort a List

Method 1:

```
arr=[4,2,5,1,3]
arr.sort()
print(arr)
```

Output:

[1, 2, 3, 4, 5]

Method 2:

```
arr=[4,2,5,1,3]
print(sorted(arr))
```

25. Merge Two Lists

Method 1:

```
a=[1,2,3]
b=[4,5,6]
print(a+b)
```

Output:

[1, 2, 3, 4, 5, 6]

Method 2:

```
a=[1,2,3]
b=[4,5,6]
a.extend(b)
print(a)
```

26. Find the Intersection of Two Lists

Method 1:

```
a=[1,2,3,4]
b=[3,4,5,6]
print(list(set(a)&set(b)))
```

Output:

[3, 4]

Method 2:

```
a=[1,2,3,4]
b=[3,4,5,6]
c=[]
for num in a:
    if num in b:
    c.append(num)
print(c)
```

27. Find Minimum and Maximum Elements in an Array

Method 1:

```
arr=[10,20,5,40,15]
```

```
print("Minimum:",min(arr))
print("Maximum:",max(arr))
```
Output:
Minimum: 5
Maximum: 40

Method 2:
```
arr=[10,20,5,40,15]
minimum=arr[0]
maximum=arr[0]
for num in arr:
    if num<minimum:
        minimum=num
    if num>maximum:
        maximum=num
print("Minimum:",minimum)
print("Maximum:",maximum)
```

28. Find Missing Number in a List

Method 1:
```
arr=[1,2,3,5]
n=5
print(n*(n+1)//2-sum(arr))
```
Output:
4

Method 2:
```
arr=[1,2,3,5]
for num in range(1,6):
    if num not in arr:
        print(num)
```

29. Count Frequency Using Dictionary

Method 1:
```
arr=[1,2,2,3,2,4]
freq={}
for num in arr:
    freq[num]=freq.get(num,0)+1
print(freq)
```
Output:
{1: 1, 2: 3, 3: 1, 4: 1}

Method 2:
```
arr=[1,2,2,3,2,4]
for num in set(arr):
    print(num,":",arr.count(num))
```
Output:
1 : 1
2 : 3
3 : 1
4 : 1

30. Dictionary Key-Value Swap

Method 1:
```
d={"a":1,"b":2,"c":3}
new={}
for key,value in d.items():
    new[value]=key
print(new)
```
Output:
{1: 'a', 2: 'b', 3: 'c'}

Method 2:
```
d={"a":1,"b":2,"c":3}
print({value:key for key,value in d.items()})
```

31. Sort a Dictionary by Value

```
d={"a":30,"b":10,"c":20}
print(dict(sorted(d.items(),key=lambda item:item)))
```

Output:

{'b': 10, 'c': 20, 'a': 30}

32. Set Operations

Method 1:

```
a={1,2,3,4}
b={3,4,5,6}
print("Union:",a|b)
print("Intersection:",a&b)
print("Difference:",a-b)
```

Output:

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference: {1, 2}

Method 2:

```
a={1,2,3,4}
b={3,4,5,6}
print("Union:",a.union(b))
print("Intersection:",a.intersection(b)
)
print("Difference:",a.difference(b))
```

33. Convert List to Tuple

```
arr=[1,2,3,4,5]
print(tuple(arr))
```

Output:

(1, 2, 3, 4, 5)

34. Multiplication Table

```
n=5
for i in range(1,11):
    print(n,"x",i,"=",n*i)
```

Output:

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

35. Sum of First N Numbers

Method 1:

```
n=5
print(n*(n+1)//2)
```

Output:

15

Method 2:

```
n=5
total=0
for i in range(1,n+1):
    total+=i
print(total)
```

36. Factorial

Method 1:

```
from math import factorial
print(factorial(5))
```

Output:

120

Method 2:

```
n=5
fact=1
for i in range(1,n+1):
```

```
    fact*=i
print(fact)
```

37. Function with Arguments

```
def add(a,b):
    return a+b
print(add(10,20))
```

Output:

30

38. Lambda Function

```
add=lambda a,b:a+b
print(add(10,20))
```

Output:

30

39. Recursive Factorial

```
def fact(n):
    if n==1:
        return 1
    return n*fact(n-1)
print(fact(5))
```

Output:

120

---

40. Recursive Fibonacci

```
def fib(n):
    if n<=1:
        return n
    return fib(n-1)+fib(n-2)
for i in range(6):
    print(fib(i),end=" ")
```

Output:

0 1 1 2 3 5

41. Prime Number

```
n=7
count=0
for i in range(1,n+1):
    if n%i==0:
        count+=1
if count==2:
    print("Prime")
else:
    print("Not Prime")
```

Output:

Prime

42. Armstrong Number

```
n=153
total=sum(int(i)**3 for i in str(n))
if total==n:
    print("Armstrong")
else:
    print("Not Armstrong")
```

Output:

Armstrong

43. Fibonacci Series

```
n=5
a=0
b=1
for i in range(n):
    print(a,end=" ")
    a,b=b,a+b
```

Output:

0 1 1 2 3

44. GCD (Greatest Common Divisor)

```
import math
print(math.gcd(12,18))
```

Output:

6

45. LCM (Least Common Multiple)

```
import math
a=12
b=18
print((a*b)//math.gcd(a,b))
```

Output:

36

46. Perfect Number

```
n=28
total=0
for i in range(1,n):
    if n%i==0:
        total+=i
if total==n:
    print("Perfect Number")
else:
    print("Not Perfect Number")
```

Output:

Perfect Number

47. Count Digits in a Number

```
n=12345
print(len(str(n)))
```

Output:

5

48. Sum of Digits of a Number

Method 1:

```
n=1234
print(sum(int(i) for i in str(n)))
```

Output:

10

Method 2:

```
n=1234
total=0
for digit in str(n):
    total+=int(digit)
```

```
print
(total)
```

49. Linear Search

```
arr=[10,20,30,40,50]
key=30
if key in arr:
   print("Found")
else:
   print("Not Found")
```

Output:

Found

50. Binary Search

```
arr=[10,20,30,40,50]
key=30
low=0
high=len(arr)-1
while low<=high:
   mid=(low+high)//2
   if arr[mid]==key:
      print("Found")
      break
   elif arr[mid]<key:
      low=mid+1
   else:
      high=mid-1
```

Output:

Found

51. Bubble Sort

Method 1:

```
arr=[5,3,8,4,2]
n=len(arr)
for i in range(n):
   for j in range(n-i-1):
      if arr[j]>arr[j+1]:
         arr[j],arr[j+1]=arr[j+1],arr[j]
print(arr)
```

Output:

[2, 3, 4, 5, 8]

Method 2:

```
arr=[5,3,8,4,2]
print(sorted(arr))
```

Output:

[2, 3, 4, 5, 8]

52. Read a File

Method 1:

```
file=open("demo.txt","r")
print(file.read())
file.close()
```

Output:

Hello World

Method 2:

```
with open("demo.txt","r") as file:
    print(file.read())
```

Output:

Hello World

53. Try-Except Example

```
try:
    a=10
    b=0
    print(a/b)
except:
    print("Cannot divide by zero")
```

Output:

Cannot divide by zero

54. Find Duplicates in a List

```
arr=[10,20,10,30,20,40]
dup=[]
for num in arr:
    if arr.count(num)>1 and num not in dup:
        dup.append(num)
print(dup)
```

Output:

[10, 20]

55. Find the Most Frequent Element

```
arr=[1,2,2,3,2,4]
print(max(arr,key=arr.count))
```

Output:

2

56. Right Triangle Star Pattern

```
n=5
for i in range(1,n+1):
    print("*"*i)
```

Output:

```
*
**
***
****
*****
```

57. Inverted Right Triangle Pattern

```
n=5
for i in range(n,0,-1):
    print("*"*i)
```

Output:

```
*****
****
***
**
```

```
*
```

58. Pyramid Star Pattern

```
n=5
for i in range(n):
    print(" "*(n-i-1)+"*"*(2*i+1))
```

Output:

```
    *
   ***
  *****
 *******
*********
```

59. Inverted Pyramid Pattern

```
n=5
for i in range(n):
    print(" "*i+"*"*(2*(n-i)-1))
```

Output:

```
*********
 *******
  *****
   ***
    *
```

60. Diamond Pattern

```
n=5
for i in range(n):
    print(" "*(n-i-1)+"*"*(2*i+1))
for i in range(n-2,-1,-1):
    print(" "*(n-i-1)+"*"*(2*i+1))
```

Output:

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

61. Number Triangle Pattern

```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end="")
    print()
```

Output:

```
1
12
123
1234
12345
```

62. Floyd's Triangle

```
n=4
num=1
```

```
for i in range(1,n+1):
    for j in range(i):
        print(num,end=" ")
        num+=1
    print()
```

Output:

```
1
2 3
4 5 6
7 8 9 10
```

63. Pascal's Triangle

```
n=5
for i in range(n):
    num=1
    print(" "*(n-i),end="")
    for j in range(i+1):
        print(num,end=" ")
        num=num*(i-j)//(j+1)
    print()
```

Output:

```
     1
    1 1
   1 2 1
  1 3 3 1
 1 4 6 4 1
```

64. Alphabet Triangle Pattern

```
n=5
for i in range(1,n+1):
    for j in range(i):
        print(chr(65+j),end="")
    print()
```

Output:

```
A
AB
ABC
ABCD
ABCDE
```

65. Hollow Square Pattern

```
n=5
for i in range(n):
    for j in range(n):
        if i==0 or i==n-1 or j==0 or j==n-1:
            print("*",end=" ")
        else:
            print(" ",end=" ")
    print()
```

Output:

```
* * * * *
*       *
*       *
*       *
```

* * * * *

66. Hollow Rectangle Pattern

```
r=4
c=6
for i in range(r):
    for j in range(c):
        if i==0 or i==r-1 or j==0 or
j==c-1:
            print("*",end=" ")
        else:
            print(" ",end=" ")
    print()
```

Output:

```
* * * * * *
*         *
*         *
* * * * * *
```